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CANDIDATE
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CHEMISTRY

Grade 12

Paper 1 Multiple Choice

May 2020

1 hour

Candidates answer on the Question Paper.

Additional Materials: Calculator
 Data Booklet

12CHEM/01

READ THESE INSTRUCTIONS FIRST

Write your centre number and candidate number in the spaces at the top of the page.

Write in dark blue or black pen.

Do not use staples, paper clips, glue or correction fluid.

DO **NOT** WRITE IN ANY BARCODES.

There are **forty** questions on this paper.

Answer **all** questions.

For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct by putting a tick (✓) in the correct box.

Each correct answer will score one mark.

Any rough working can be done in this booklet.

The total number of marks for this paper is 40.

Answer all questions in English.

For Examiner's Use

Total:

This document consists of **16** printed pages.



Section A

For
Examiner's
Use

For each question there are four possible answers, **A**, **B**, **C** and **D**. Choose the one you consider to be correct by putting a tick (✓) in the correct box.

1 What is the electron configuration of one atom of bromine?

A $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^5$

B $1s^2 2s^2 2p^6 3s^2 3d^6 3p^{10} 4s^2 4p^7$

C $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4d^2 4p^5$

D $1s^2 2s^2 2p^6 3s^2 3p^8 3d^8 4s^2 4p^5$

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

2 Sulfur dioxide, SO_2 , has four bonding pairs and one lone pair surrounding the central sulfur atom.

What is the shape of a sulfur dioxide molecule?

A linear

B pyramidal

C tetrahedral

D V-shape

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

3 The boiling points of two compounds are shown in the table

compound	boiling point / °C
butane	-0.5
butan-1-ol	117

Which statement explains the difference in boiling points of the compounds?

A Butane contains less electrons than butan-1-ol.

B Butane does not have hydrogen bonds but butan-1-ol does.

C Butane is a non-polar molecule but butan-1-ol is polar molecule.

D Butane has a smaller molar mass than butan-1-ol.

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

- 4 Which row of the table shows the hybridisation of the carbon atom and the number of sigma (σ) bonds in carbon dioxide?

	type of hybrid orbitals	number of sigma bonds
A	sp^2	one
B	sp^2	two
C	sp	one
D	sp	two

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

- 5 Silicon dioxide, SiO_2 , has a macromolecular covalent lattice.

What type of bonding exists between silicon atoms and oxygen atoms in this structure?

- A** covalent
B ionic
C metallic
D van der Waals'

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

- 6 The infra-red spectrum of an organic acid shows absorption at 1700cm^{-1} and between $3200\text{--}3600\text{ cm}^{-1}$.

Which statement is correct about these absorptions?

- A** The absorption of infra-red causes increased vibrations in the O-H and C=O bonds.
B The absorption of infra-red causes electronic transitions in the O-H and C=O bonds.
C The absorption of infra-red causes rotation in the O-H and C=O bonds.
D The absorption of infra-red breaks the O-H and C=O bonds.

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

7 Who is the author of spiral system of the chemical elements?

- A Lothar Meyer
- B Moseley
- C Newlands
- D Shankurtua

A ☐ B ☐ C ☐ D ☐

[1]

8 What property decreases as atomic number increases in Group 1?

- A atomic mass
- B atomic radius
- C density
- D ionisation energy

A ☐ B ☐ C ☐ D ☐

[1]

9 Sulfuric acid is manufactured by the Contact process.

Which statement about the Contact process is correct?

- A Nitrogen(I) oxide is used as a catalyst.
- B It uses a temperature of 500°C.
- C When sulfur oxide is absorbed, water is used.
- D It uses a pressure of 200 atmospheres.

A ☐ B ☐ C ☐ D ☐

[1]

- 10 The combustion of coal produces a gas containing sulfur.

Which environmental problem does this gas cause?

- A acid rain
B eutrophication
C greenhouse effect
D ozone depletion

A ☐ B ☐ C ☐ D ☐

[1]

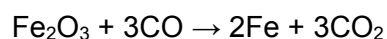
- 11 Which row of the table shows a reaction that uses a homogenous catalyst?

	reaction	catalyst
A	$2\text{N}_2\text{O(g)} \rightarrow 2\text{N}_2\text{(g)} + \text{O}_2\text{(g)}$	$\text{Cl}_2\text{(g)}$
B	$2\text{H}_2\text{O}_2\text{(aq)} \rightarrow 2\text{H}_2\text{O(l)} + \text{O}_2\text{(g)}$	$\text{MnO}_2\text{(s)}$
C	$\text{N}_2\text{(g)} + 3\text{H}_2\text{(g)} \rightarrow 2\text{NH}_3\text{(g)}$	Fe(s)
D	$2\text{CO(g)} + 2\text{NO(g)} \rightarrow 2\text{CO}_2\text{(g)} + \text{N}_2\text{(g)}$	Pt(s)

A ☐ B ☐ C ☐ D ☐

[1]

- 12 Iron is produced in the reduction of iron(III) oxide by carbon monoxide.



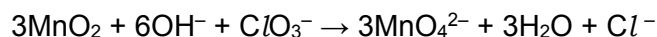
What is the atom economy for the production of iron by this method?

- A 70.0
B 54.1
C 45.9
D 35.0

A ☐ B ☐ C ☐ D ☐

[1]

- 13 The equation for a redox reaction is shown:



How many electrons are gained or lost by manganese during the reaction as shown?

- A** 1
B 2
C 4
D 6

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

- 14 Use of the Data Booklet is relevant to this question.

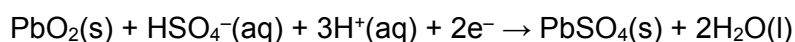
Which letter shows the order of metals from the weakest to the strongest reducing agent?

- A** Ag Fe Cu Na
B Na Fe Cu Ag
C Ag Cu Fe Na
D Fe Na Ag Cu

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

- 15 The half-equations at each electrode during the discharge of a lead-acid battery are shown.



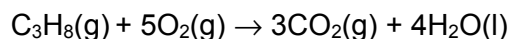
Which products are formed when a lead-acid battery is recharged?

- A** $\text{PbSO}_4(\text{s}) + \text{H}_2\text{O(l)}$
B $\text{Pb(s)} + \text{PbO}_2(\text{s}) + \text{H}^+(\text{aq}) + \text{HSO}_4^-(\text{aq})$
C $\text{Pb(s)} + \text{HSO}_4^-(\text{aq}) + \text{PbSO}_4(\text{s}) + \text{H}_2\text{O(l)}$
D $\text{PbSO}_4(\text{s}) + \text{H}^+(\text{aq}) + \text{PbO}_2(\text{s}) + \text{HSO}_4^-(\text{aq})$

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

- 16 The equation for the combustion of propane, $\text{C}_3\text{H}_8(\text{g})$, is shown:



Some standard enthalpy changes of formation are shown

compound	enthalpy change of formation, ΔH_f^θ / kJ mol^{-1}
$\text{CO}_2(\text{g})$	-393
$\text{H}_2\text{O}(\text{l})$	-286
$\text{C}_3\text{H}_8(\text{g})$	-104

What is the standard enthalpy of combustion, ΔH_c^θ , of propane in kJ mol^{-1} ?

- A -2427 kJ mol^{-1}
 B -2219 kJ mol^{-1}
 C 575 kJ mol^{-1}
 D 2219 kJ mol^{-1}

A ☐ B ☐ C ☐ D ☐

[1]

- 17 Which row of the table shows a reaction that is **never** spontaneous?

	ΔH^θ	ΔS^θ	T
A	negative	negative	low
B	positive	positive	high
C	positive	negative	low
D	negative	positive	low

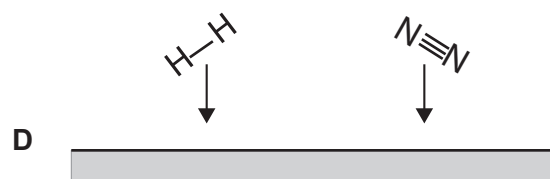
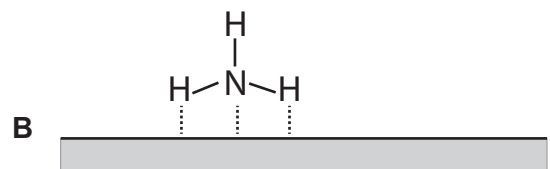
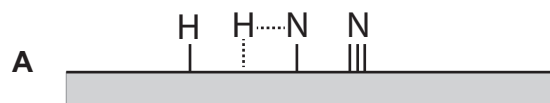
A ☐ B ☐ C ☐ D ☐

[1]

18 The diagrams show the mechanism of heterogeneous catalysis.

Which diagram shows the process of new bonds formation?

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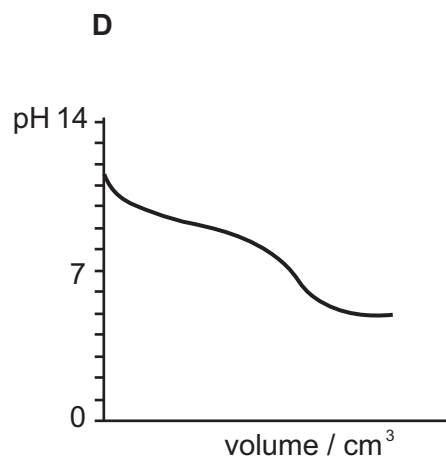
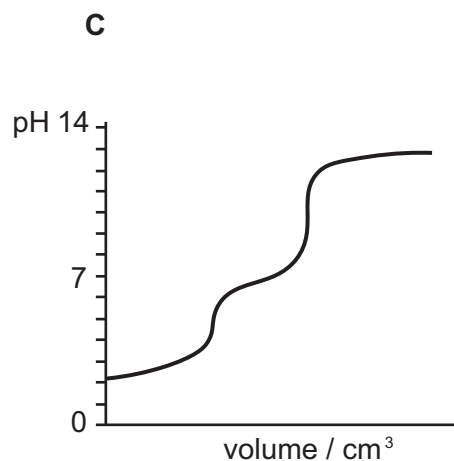
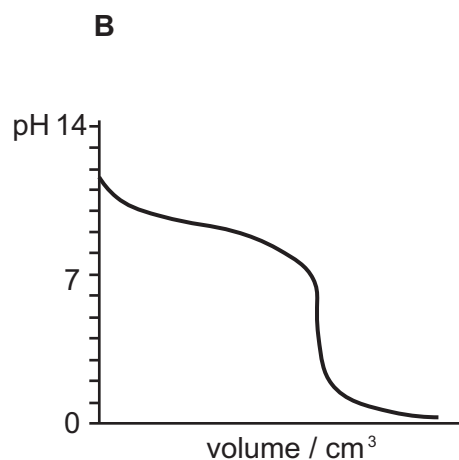
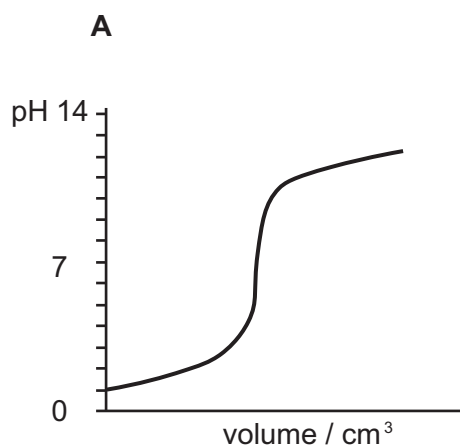
A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

- 19 A student adds aqueous sodium hydroxide slowly from a burette into a flask containing propanedioic acid, $\text{HOOCCH}_2\text{COOH}$.

For
Examiner's
Use

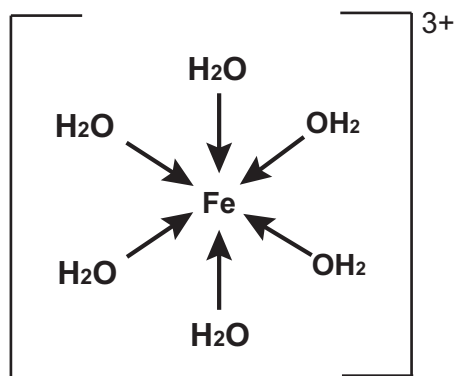
Which curve shows the change in the pH of the contents of the flask as the sodium hydroxide is added?



A ☐ B ☐ C ☐ D ☐

[1]

20 The diagram shows the structure of a complex ion.



Which statement about this ion is correct?

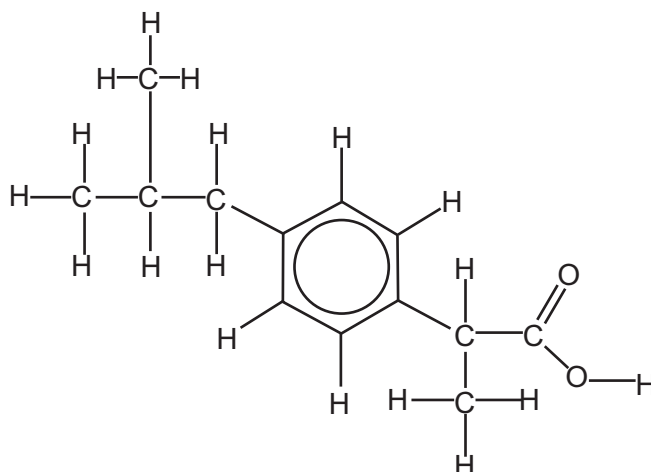
- A It produces a brown precipitate with aqueous sodium hydroxide.
- B It produces a green precipitate with aqueous ammonia.
- C The Fe^{3+} has donated an electron pair.
- D The Fe^{3+} has accepted a proton.

A ☐ B ☐ C ☐ D ☐

[1]

21 Ibuprofen is used in medicine as a painkiller.

The structural formula of Ibuprofen is shown.



Which functional groups does the ibuprofen molecule present?

- A amine, ester
- B ketone, nitrile
- C aldehyde, alcohol
- D phenyl, carboxyl

A ☐ B ☐ C ☐ D ☐

[1]

For
Examiner's
Use

- 22 Propene, C_3H_6 , reacts with water in the presence of phosphoric acid at a temperature of $300^\circ C$.

What is the mechanism of this reaction?

- A nucleophilic substitution
- B nucleophilic addition
- C electrophilic addition
- D electrophilic substitution

A ☐ B ☐ C ☐ D ☐

[1]

- 23 Which of the statements about esters is correct?

- A The esterification of an alcohol with a carboxylic acid will go to completion.
- B Esters are formed **only** by carboxylic acids.
- C An alcohol is eliminated during the hydrolysis of an ester.
- D Ethanyl chloride reacts with ammonia to make an ester.

A ☐ B ☐ C ☐ D ☐

[1]

- 24 Which property about polyvinyl chloride, PVC, is correct?

- A PVC conducts electricity.
- B PVC does not melt at high temperatures.
- C PVC is solid and rigid at room temperature, but it can be plasticised.
- D PVC corrodes easily in the presence of water and air.

A ☐ B ☐ C ☐ D ☐

[1]

25 Amines behave both as a nucleophile and as a base.

Which row in the table is correct?

	behaves as a nucleophile	behaves as a base
A	accepts an electron pair	accepts a proton
B	decomposes to form ammonia	turns moist red litmus blue
C	reacts with an acid (acyl) chloride to form an amide	accepts an electron pair
D	donates an electron pair to an electron deficient centre	reacts with an acid to form a salt

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

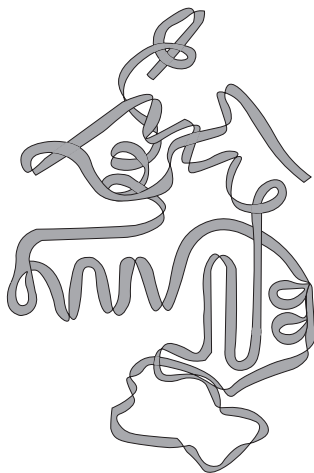
26 Which statement explains why phenylamine, $\text{C}_6\text{H}_5\text{NH}_2$, is a weaker base than ethylamine $\text{C}_2\text{H}_5\text{NH}_2$?

- A** Phenylamine has a greater negative charge than ethylamine.
- B** The lone pair of the NH_2 group is delocalised into the benzene ring.
- C** The benzene ring is electron releasing.
- D** Phenylamine is more covalent than ethylamine.

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

27 Which types of protein structure are shown in the figure?



- A** primary and secondary
- B** primary and quaternary
- C** secondary and tertiary
- D** tertiary and quaternary

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

- 28 Which row in the table shows a base pair found in DNA and the number of hydrogen bonds between the base pair?

For
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Use

	base pair	number of hydrogen bonds
A	adenine-thymine	three
B	cytosine-thymine	two
C	adenine-guanine	two
D	cytosine-guanine	three

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

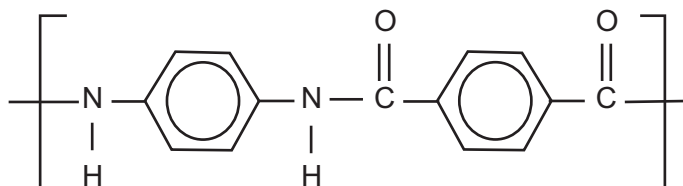
- 29 What is the concentration of hydrogen ions in 0.15 mol dm^{-3} of aqueous ethylamine, $\text{C}_2\text{H}_5\text{NH}_2$? The pK_b for ethylamine is 3.27.

- A** $11.95 \text{ mol dm}^{-3}$
B $8.97 \times 10^{-3} \text{ mol dm}^{-3}$
C $5.37 \times 10^{-4} \text{ mol dm}^{-3}$
D $1.11 \times 10^{-12} \text{ mol dm}^{-3}$

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

- 30 The repeating unit for Kevlar is shown.



Which products are formed when Kevlar is hydrolysed?

- A** benzene-1,6-dicarboxylic acid and 1,6-diaminobenzene
B benzene-1,4-dicarboxylic acid and 1,4-diaminobenzene
C 1,4-diaminohexane and hexanedioic acid
D 1,6-diaminohexane and hexanedioic acid

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

Section B

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Use

For each of the questions in this section, one or more of the three numbered statements **1** to **3** may be correct.

Decide whether each of the statements is correct (you may find it helpful to put a tick against the statements that you consider to be correct).

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct.	1 and 2 only are correct.	2 and 3 only are correct.	1 only is correct.

No other combination of statements is used as a correct response.

31 Which metal(s) react(s) with gaseous water at high temperature to form a metal oxide and hydrogen?

1 Mg

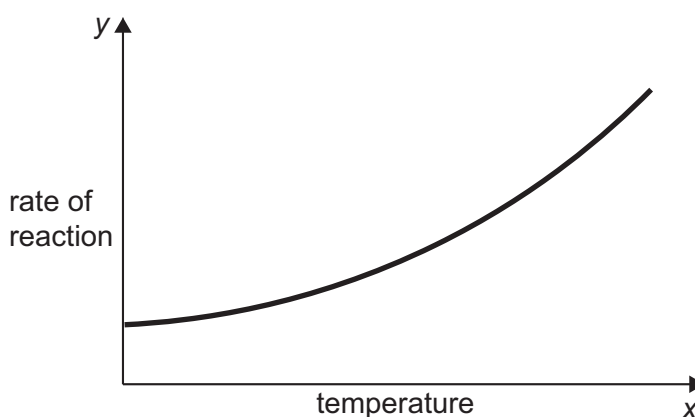
2 Fe

3 Cu

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

32 The change in rate of reaction with temperature is shown in the graph.



Which statement(s) is/are correct?

1 The rate of reaction increases as the temperature increases.

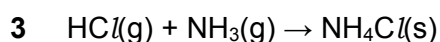
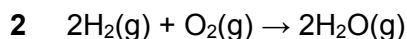
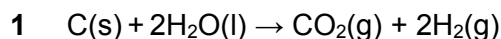
2 As the rate of reaction increases the time of complete reaction decreases.

3 The graph shows that the reaction is second order.

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

33 Which reaction(s) represent(s) an overall increase in the entropy of the system?



A ☐ B ☐ C ☐ D ☐

[1]

34 Use of the Data Booklet is relevant to this question.

Which wavenumber range(s) occur(s) in the infra-red absorption spectrum of phenylamine, $\text{C}_6\text{H}_5\text{NH}_2$?

1 1450-1650 cm^{-1}

2 3000-3100 cm^{-1}

3 3300-3500 cm^{-1}

A ☐ B ☐ C ☐ D ☐

[1]

35 Which of the alcohols can be oxidised to a ketone?

1 3-methylpentan-2-ol

2 2-methylpentan-3-ol

3 2,3-dimethylbutan-1-ol

A ☐ B ☐ C ☐ D ☐

[1]

36 Which statement(s) about the thermal stability of the compounds of Group 14 elements is/are correct?

1 carbon monoxide is more thermally stable than germanium(II) oxide

2 lead(II) oxide is more thermally stable than lead(IV) oxide

3 carbon tetrachloride is more thermally stable than tin(IV) chloride

A ☐ B ☐ C ☐ D ☐

[1]

37 Which substances are formed when concentrated sulfuric acid is added to a solid sample of sodium iodide?

- 1 sulfur dioxide
- 2 hydrogen sulfide
- 3 iodine

A ☐ B ☐ C ☐ D ☐

[1]

38 Which spin-spin splitting(s) is/are possible in the ^1H -NMR spectrum of 1-chloro-2-methylpropane?

- 1 doublet
- 2 multiplet
- 3 singlet

A ☐ B ☐ C ☐ D ☐

[1]

39 Propanoic acid reacts with sodium carbonate.

Which substance(s) is/are formed in this reaction?

- 1 sodium chloride
- 2 carbon dioxide
- 3 water

A ☐ B ☐ C ☐ D ☐

[1]

40 Use of the Data Booklet is relevant to this question.

In which electrochemical cell(s) do/does reaction(s) occur spontaneously under standard conditions?

- 1 $\text{H}_2(\text{g}) \mid 2\text{H}^+(\text{aq}) \parallel \text{Mg}^{2+}(\text{aq}) \mid \text{Mg}(\text{s})$
- 2 $\text{V}(\text{s}) \mid \text{V}^{2+}(\text{aq}) \parallel \text{Sn}^{4+}(\text{aq}) \mid \text{Sn}^{2+}(\text{aq})$
- 3 $2\text{Cl}^-(\text{aq}) \mid \text{Cl}_2(\text{g}) \parallel \text{MnO}_4^-(\text{aq}) \mid \text{Mn}^{2+}(\text{aq})$ under acidic conditions

A ☐ B ☐ C ☐ D ☐

[1]

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